Energy Wave

The power supply for humanoid robots.

SEED ROUND — RESEARCH VENTURE

energywave.ai

CONFIDENTIAL — FORECAST — FOR DISCUSSION

ONE IMPORTANT TRUTH

> *“What important truth do very few people agree with you on?”*

The last forty years compounded bits, not watts — and the inflection point to the physical-world future is energy density at the point of use.

Look around: the world software built is everywhere; the world it was supposed to power is not. EnergyWave exists to determine whether the next step in portable power is physically achievable — and to own the IP if it is.

The objective is not to prove a theory. It is to discover whether reality supports one. The seed round funds that discovery to its first decision, and pays for the next decision only if the first one says go.

THE PROBLEM

The energy layer never had its Moore’s Law.

Information technology rode decades of exponential gains. The power that drives the physical world — motors, robots, vehicles — did not. The gap between the two curves is the opportunity.

Lithium-ion sits at a practical 150–250 Wh/kg against a 400–500 Wh/kg theoretical range; industry planning assumes about 4 % a year, a doubling in roughly eighteen years. Platform makers are engineering around it — swappable packs, docks, inductive pads in the feet — not through it. *Attributed published figures (DoW white paper v2.0, sources [5]–[7]); not company measurements.*

~10^6×+

Growth in compute per dollar since 1990

~3–4×

Improvement in portable energy density over the same period

Directional, not to scale — orders of magnitude illustrate the divergence.

THE CAPABILITY

A humanoid is a power problem.

To sense and move like a person, a humanoid must run human-grade systems at once — not a few proximity sensors and a battery clock.

Sight & sound

Cameras and microphones.

Smell & taste

Gas-sensor and electrochemical arrays — electronic nose and tongue.

Touch

Tactile arrays and electronic skin, at nerve-level granularity.

Motion

Dense actuators, at muscle-level granularity.

Whether that simultaneous load exceeds what a lithium pack delivers is a question to quantify for a defined platform — the power-budget specification (a capital-readiness deliverable, drafted; platform inputs specified at the design review) is built to answer it. **EnergyWave sets out to build the power supply for that load. Nothing else.**

WHERE WE ARE

The bridge before this round — and what it had to deliver.

The capital-readiness bridge — a $60,000 SAFE on the October materials; no SAFE executed and no outside funding raised; the current amount and instrument set with counsel at close — is to buy three months and three objectives. Capital Readiness may be combined with this round. Each objective is shown as it stands today.

#	CAPITAL-READINESS OBJECTIVE	EVIDENCE THAT CLOSES IT	TODAY
1	Form both entities; IP strategy with K&L Gates; initial provisional protection	Delaware certificate and file number; EIN; a coversheet provisional filed in the inventors' names; 501(c)(3) formation documents	Formation in progress with K&L Gates
2	Technical review of the first module; humanoid power-budget specification	A characterization plan with pre-registered pass / kill criteria; the power-budget specification — both drafted; inputs specified at kickoff	Drafted
3	Measurable go / no-go criteria; the next round prepared	Gate criteria published before any result; this round's amount, milestones and use of funds sized from the forecast model	Drafted — projected

Capital raised: **no outside debt or equity, no SAFE executed, no outside funding raised** (15 Oct 2026); the October competition entry recorded approximately $10,000 including founder contributions. Entity: EnergyWave, Inc., Delaware — **formation in progress with K&L Gates LLP**; formation is a condition of closing.

THE RESEARCH QUESTION

Three threads. One protocol. No thread feeds another.

Do engineered structures — a conditioned solid-state magnetic core, a Casimir cavity, a metal–insulator–metal junction — produce any repeatable, artifact-free electrical output whose energy accounting cannot be closed by known sources? Expected answer on the mainstream prior: no. The company exists to obtain that answer to a standard that survives review, and to publish it either way.

THREAD A

Founder's solid-state module

Floyd Sweet's vacuum triode amplifier — the named first module and, since 15 October 2026, **the lead research path**. No specimen selected, no hardware ordered, no measured result. Characterization plan drafted; the specimen, magnetization, winding axis and conditioning procedure are specified by the inventor at kickoff.

THREAD B

Independent replication

Pre-registered replication of Moddel et al. 2021 — MIM diode under a Casimir cavity, cavity / no-cavity twins from one wafer, blinded. Published claim 1.4 pW, one lab, unreplicated; Maclay 2024 predicts the opposite sign. *Claimed, not replicated.*

THREAD C

The measurement platform

Shielded isothermal enclosure, calorimetric accounting, lock-in measurement to the picowatt floor, blinded handling, open data. Published as a methods paper before any result. Residual value if every thread fails.

Sweet solid-state is the lead path — the founder's decision of 15 October 2026. A selected lead path is not a measured result or an operational schedule. Threads B and C remain distinct workstreams under the same protocol; the order on the shared bench and the measurement lead are set at kickoff with the principal investigator.

THE GATE PROTOCOL

Research first, claims last.

GATE	QUESTION	PASS	KILL
Gate 1 Characterize	Can the reported effects be reproduced, and are they artifacts?	Repeatable output above the fully characterized noise and artifact floor at ≥ 5σ in blinded runs, surviving every control.	Nothing above the floor, or output explained by a known source. Published with data.
Gate 2 Conservation test	If something is there, where does the energy come from?	Net output whose closed-system balance cannot be closed by known sources, replicated by a second laboratory contracted in advance.	The balance closes, or replication fails. Effect retired publicly.
Gate 3 Extract or retire	Can it be engineered?	Independently validated, repeatable performance under third-party observation — the first point at which EnergyWave would ever state a number.	Scaling, lifetime or stability fails.

$E_{\text{all inputs}} = E_{\text{load}} + E_{\text{heat and other outputs}} + \Delta U_{\text{device}}$. DC input is $\int v \cdot i \, dt$, never average V × average I; probe deskew in a high-Q circuit can turn +50 mW of dissipation into a false −75 mW “generation” reading at 1 MHz. A null result bounds the tested configuration at its demonstrated sensitivity; it does not identify an origin.

WHAT THE SEED FUNDS

Capital follows evidence. Tranche A at close; B and C in escrow.

TRANCHE	FUNDS	RELEASED ON	PROJECTED CUMULATIVE NEED (FORECAST MODEL V1.2, BASE)
A · at close	Gate 1 — commission the artifact-exclusion bench and publish its error budget (C); run the pre-registered replication (B); characterize the founder's module under the same protocol (A); hire the principal investigator — a condition of Tranche A.	Close. Projected Tranche A ≈ $7.0M.	$6,987,330 by month 9
B · escrowed	Gate 2 — closed-system calorimetry; mechanism discrimination; independent replication at a second laboratory contracted before Gate 2 begins.	Written Gate-1 report, advisory-board certification, publication. Projected Tranche B ≈ $6.7M.	$13,719,010 by month 21 (+ $6,731,680)
C · escrowed	Gate 3 — engineering characterization, lifetime and stability, third-party validation; the anti-tamper paperweight test.	Written Gate-2 report, certification, publication. Projected Tranche C ≈ $8.4M.	$22,143,610 by month 36 (+ $8,424,600)

Right-hand column: **projected cumulative need from the forecast model v1.2 (16 October 2026), Base scenario** — projections on assumptions to be validated by finance, including two theoretical-physicist agreements modelled at $500k per year *each* and the co-founder at $500,000; tranche sizes are set with counsel at close. Investors receive the same gate reports the public does. Unspent escrow is returned on a kill.

USE OF FUNDS — 36-MONTH PLAN, AS THE FORECAST MODEL PROJECTS IT (V1.2, BASE SCENARIO)

Where the money would go, as projected.

CATEGORY (BASE SCENARIO, PROJECTED)	YEAR 1	YEARS 2–3	36 MONTHS
Payroll — 18 roles, five applied physicists, the co-founder at $500,000	2,830,708	5,904,800	8,735,508
Consulting — incl. two theoretical physicists at $500k/yr each	976,667	2,120,000	3,096,667
Equipment — instrumentation, additive manufacturing	1,255,000	140,000	1,395,000
Facilities — modelled as 6,000 sq ft near UC Irvine (assumption)	636,500	636,000	1,272,500
Research · Legal · G&A · Finance · Operations	1,521,333	2,432,000	3,953,333
Plan spend	**7,220,208**	**11,232,800**	**18,453,008**
20 % reserve (on the whole plan)			3,690,602
Projected raise including reserve			**22,143,610**

ORIGIN OF EVERY NUMBER HERE

Projections of the forecast model v1.2 (16 October 2026; Base scenario, recalculated in full) on forecast assumptions to be validated by finance — not raise commitments. One-time spend $2,145,000; recurring $16,308,008; monthly burn at the model's snapshot months 6 / 12 / 24 / 36: $462,200 / $468,033 / $468,033 / $468,033 (point-month outputs, not an average). Lean $15.8M plan / $19.0M raise; Full $22.3M / $26.8M.

BEFORE ANY OF IT IS A COMMITMENT

Founder salary, engineer bases, rent, build-out, the instrumentation package, the additive-manufacturing system, insurance, recruiting, and the consulting-agreement question — validated by finance before close. Planned office: Irvine, near UC Irvine and the National Fuel Cell Research Center; site and lease to be negotiated.

MILESTONES

Three decisions, on the model’s clock.

9 MONTH 9 · GATE 1

Characterize

- Measurement platform commissioned on known-null samples; error budget published for external review
- Thread B replication run blinded; thread A characterized under the same protocol
- Gate-1 report; advisory-board certification; publication — go, or wind-down with escrow returned

21 MONTH 21 · GATE 2

Conservation test

- Closed-system calorimetry on anything that survived Gate 1
- Mechanism discrimination against known-source models
- Second-laboratory replication, contracted before the gate begins — go, or the effect is retired publicly

36 MONTH 36 · GATE 3

Extract or retire

- Engineering characterization; lifetime and stability
- Third-party validation under observation — the first performance statement, if any
- Anti-tamper paperweight test passed; Series A becomes possible

Gate months are the forecast model’s assumptions (plan start 1 January 2027). The August 2026 white paper phases them as 0–6, 6–18 and 18–36 months. Same three decisions.

TEAM & COUNSEL

Built to be checked, not just believed.

Ian McLean Founder & CEO

Founded Statpedia; FDIC (2008 crisis); Wolfram Research; Avnet venture tech — early Anduril, Wrap (NASDAQ: WRAP). DoW white paper author; NDA consulting, U.S. intelligence org.

Chip Story Co-Founder · Physicist

B.S. geological engineering (South Dakota School of Mines); M.S. geophysics (Colorado School of Mines); 50+ years in energy exploration — published researcher. Director, Zero Point Energy Project Group.

Jack Sarfatti Team member — theoretical physicist

B.A. physics, Cornell — trained under Hans Bethe; Ph.D. physics, UC Riverside. Birkbeck College with David Bohm; ICTP Trieste with Abdus Salam. Fundamental Fysiks Group. Team member. A critic of the premise, which the company treats as a credibility asset.

Alan Crunkleton Technical & financial advisor

MIT Ph.D. (mech. eng., 1987); co-inventor of the Boreas cycle; co-founder and president of Boreas, Inc.; R&D 100 award (1994). 35+ years, concept to commercialization.

Current core team: Ian McLean and Chip Story. Team members: Jack Sarfatti; Mike Wentz (role to be announced). Advisor: Alan Crunkleton. Planned from the Seed Round: Arath Acosta — Design Engineer. **K&L Gates LLP — counsel:** Brian Novak; Dennis Majewski. Biographical lines are the individuals' own statements as carried on the October materials; details in *Team and Counsel*. Team and advisor agreements are set with counsel on formation.

Why physicists expect "no", and what the structure does about it.

The physics prior

The vacuum is the ground state; the Casimir force is conservative; steady work from one equilibrium reservoir is a perpetual-motion machine of the second kind. Even the best single-use Casimir stack holds ≈ 2 J/kg against ≈ 9×10^5 J/kg in the lithium cell a humanoid already carries. **Mitigation:** Gate 1 finds this out for the price of Tranche A, and publishes it.

Measurement artifacts

Probe deskew, thermal gradients, RF pickup, stored-energy depletion masquerading as generation — the history of the field. **Mitigation:** thread C exists to exclude them; error budget published; blinded runs; ≥ 5σ above a characterized floor.

Replication fails

One laboratory's result is not a result. **Mitigation:** replication contracted before each gate; a second laboratory before Gate 2; kill criteria pre-registered.

Entity, IP, agreements

Formation (in progress with K&L Gates), the provisional's successor, the review of prior agreements and the team agreements are conditions of closing. **Mitigation:** all are conditions, not aspirations; nothing closes until they do.

Key person

A founder-led research venture. **Mitigation:** the principal-investigator hire is a condition of Tranche A; the advisory board holds a veto over performance statements.

Capital

Every number in the forecast model is an assumption until finance validates it. **Mitigation:** tranches sized at close after validation; escrow returns unspent capital on a kill; investors see the same reports the public does.

EIGHT COMMITMENTS, WRITTEN IN FROM DAY ONE

The rules the money is subject to.

1. Pre-registration of every protocol, public and timestamped.
2. Null results published, with data, every time.
3. Replication budgeted and contracted before each gate, never assumed.
4. External scientific advisory board with veto over any performance statement — at least one declared critic.
5. No performance claim before Gate-3 third-party validation.
6. Capital released by gate — escrowed tranches.
7. Clean entity separation: EnergyWave, Inc. / McLean Companies LLC / GovThink.
8. Open methods: bench design and error-budget methodology published.

The two-question filter governs every document: **what is the net energy source — and where is the colder sink?** A claim that cannot answer both is documented and set down. Legitimate adjacent science — Casimir effect, Lamb shift, metamaterials, ion-wind propulsion — is never conflated with the contested leap of continuous net extraction.

Proposed structure, safety requirements and financing terms.

EnergyWave, Inc. **Delaware C corporation — formation in progress**

The operating company — the power supply for humanoid robots, nothing else. Formation in progress with K&L Gates LLP (October 2026); formation is a condition of closing. Intended to pursue strategic investment from In-Q-Tel — an intent, not a relationship.

GovThink **proposed, separate 501(c)(3) policy organization**

Proposed to make the safety and competitiveness case to government and engage on technology-transfer policy. Not yet filed; no formal tie to EnergyWave.

Safety is a design requirement. Proposed requirements, not features of any existing unit: scoped to humanoid robots only, keyed to its platform like a cartridge, designed to self-disable into an inert form if tampered with — tested at Gate 3 (the paperweight test).

TERM	VALUE
Instrument · pre-money	Set with counsel — no SAFE executed, no outside funding raised; the projected sizing is the forecast model's.
Escrow for B and C	Agent, agreement and bank confirmed directly before close; release only on gate report, certification and publication; no fee to release — a proposed structure, finalized with counsel.
Securities exemption	Rule 506(b) or 506(c), as counsel determines.

THE ASK

A seed round that buys a decision, not a three-year belief.

An investor underwrites Tranche A and Gate 1. Tranches B and C sit in escrow from the close and are released only on a written gate report, advisory-board certification and publication. “Too early for us” — the objection the company has already heard from one fund — becomes a staged option.

The amount, the pre-money and the instrument are set with counsel; the projected sizing — $22.1M over 36 months in three gate tranches — is the forecast model’s (v1.2, Base scenario), shown on slide 8 with its assumptions.

$22.1M projected

Seed · Tranche A at close · B and C escrowed

36-month plan to Gate 3 on the model’s clock

Fund the first gate.
Learn the answer either way.

Ian McLean · EnergyWave, Inc.
ian@energywave.ai · 949-280-0936 · energywave.ai

Early-stage research venture. Forward-looking statements about future technology are not proven capabilities or guarantees. Projections are forecasts of the company's model on stated assumptions, not commitments. For discussion only; not an offer to sell, or a solicitation of an offer to buy, securities.